

JOB & BATCH COSTING - CONCEPTS

1. Job Costing

- It is that form of specific order costing under which each job is treated as a cost unit and costs are accumulated and ascertained separately for each job.
- In other words, it is that form of specific order costing which applies where work is undertaken according to customer's requirement.
- It is generally used in industries where production is not on continuous basis, rather it is only when order from customers are received according to their specifications e.g. printing press, repair shop, etc.
- In this method cost of each job is computed by preparing the Job Cost Sheet.

Question – 1

The following data presented by the supervisor of a factory for a Job.

	₹ per unit
Direct material	120
Direct wages @ ₹ 4 per hour	60
(Department A-4 hrs., B-7hrs, C-2hrs & D-2hrs)	
Chargeable Expenses	<u>20</u>
Total	<u>200</u>

Analysis of the Profit and Loss Account for the year ended 31st March 2019

	₹		₹
Material used	2,00,000	Sales	4,30,000
Direct Wages:			
Dept. A	12,000		
Dept. B	8,000		
Dept. C	10,000		
Dept. D	<u>20,000</u>		
	50,000		
Special Stores Items	6,000		
Overheads:			
Dept. A	12,000		
Dept. B	6,000		
Dept. C	9,000		
Dept. D	<u>17,000</u>		
	44,000		
Gross Profit c/d	<u>1,30,000</u>		
	4,30,000		<u>4,30,000</u>
Selling Expenses	90,000	Gross Profit b/d	1,30,000

Net Profit	<u>40,000</u>	
	<u>1,30,000</u>	<u>1,30,000</u>

It is also to be noted that average hourly rates for all the four departments are similar. Required:

- Prepare a Job Cost Sheet
- Calculate the entire revised cost using the above figures as the base.
- Add 20% profit on selling price to determine the selling price.

Solution

Working Notes:

Overhead recovery rate on overall basis:

$$\text{Overhead recovery rate} = \frac{44,000}{\left(\frac{50,000}{4}\right)} = ₹ 3.52$$

Statement of calculation of recovery rates

Particulars	Working	Recovery Rate
Dept. A	$\frac{12,000}{\left(\frac{12,000}{4}\right)}$	₹ 4 per direct labour hour
Dept. B	$\frac{6,000}{\left(\frac{8,000}{4}\right)}$	₹ 3 per direct labour hour
Dept. C	$\frac{9,000}{\left(\frac{10,000}{4}\right)}$	₹ 3.60 per direct labour hour
Dept. D	$\frac{17,000}{\left(\frac{20,000}{4}\right)}$	₹ 3.40 per direct labour hour
Selling exp. As % of NFC	$\frac{90,000}{2,00,000 + 50,000 + 6,000 + 44,000} \times 100$	30% of NFC

Statement of calculation of cost and selling price of Job

Particulars	Working	Amount (₹)
Material		120
Wages		60
Chargeable expenses		20
Prime Cost		200
(+) Overheads	Dept. A = $4 \times 4.00 = 16$ Dept. B = $7 \times 3.00 = 21$ Dept. C = $2 \times 3.60 = 7.20$ Dept. D = $2 \times 3.40 = 6.80$	51

GFC\NFC		251
(+) Selling expenses	$30\% \times 251$	75.30
Total Cost		326.30
Add: Profit	$407.88 \times 20\%$	81.58
Selling price	$326.30 \div 80\%$	407.88

Question – 2

In a manufacturing company, factory overheads are charged as fixed percentage basis on direct labour and office overheads are charged on the basis of percentage of factory cost. The following information are available related to the year ending 31st March:

	Product A	Product B
Direct materials	₹ 19,000	₹ 15,000
Direct Labour	₹ 15,000	₹ 25,000
Sales	₹ 60,000	₹ 80,000
Profit	25% on cost	25% on sales price

You are required to find out:

- The percentage of factory overheads on direct labour
- The percentage of office overheads on factory cost

Solution

Let factory OH % on Direct labour = x

Let administration OH % on net factory cost = y

Statement of Cost

	Product A	Product B
Sales	60,000	80,000
Profit	12,000 $[(60,000 \times (25/125))]$	20,000 $(80,000 \times 25\%)$
Total Cost	48,000	60,000

Statement of Cost

	Product A	Product B
Direct Material	19,000	15,000
Direct labour	15,000	25,000
Prime cost	34,000	40,000
Factory OHs	150x	250x
NFC/COP/COGS	$34,000 + 150x$	$40,000 + 250x$
(+) Admin. OH	$340y + 1.5xy$	$400y + 2.5xy$
COS	$34,000 + 150x + 340y + 1.5xy$	$40,000 + 250x + 400y + 2.5xy$

$$\therefore 34,000 + 150x + 340y + 1.5xy = 48,000 \quad \text{_____ (1)}$$

$$\& 40,000 + 250x + 400y + 2.5xy = 60,000 \quad \text{_____ (2)}$$

Multiply equation (1) by 2.5 and equation (2) by 1.5 and subtract them, we get

$$85,000 + 375x + 850y + 3.75xy = 1,20,000$$

$$\pm 60,000 \pm 375x \pm 600y \pm 3.75xy = \pm 90,000$$

We get,

$$25,000 + 250y = 30,000$$

$$y = 20$$

Put value of $y = 20$ in equation (1),

$$34,000 + 150x + 340(20) + 1.5x(20) = 48,000$$

$$x = 40$$

Thus, Factory OH % on direct labour = 40% and administration OH % on factory cost = 20%

Question – 3

In a factory following the Job Costing Method, an abstract from the work-in-progress as at 30th June was prepared as under:

Job No.	Materials ₹	Direct Labour Hours	Labour ₹	Factory overheads applied
115	1,325	400 hrs	800	640
118	810	250 hrs	500	400
120	<u>765</u>	237.5 hrs	<u>475</u>	<u>380</u>
	<u>₹ 2,900</u>		<u>₹ 1,775</u>	<u>₹ 1,420</u>

Materials used in July were as follows:

Material requisition no.	Job No.	Cost (₹)
54	118	300
55	118	425
56	118	515
57	120	665
58	121	910
59	124	<u>720</u>
		<u>3,535</u>

A summary of labour hours deployed during July is as under:

Job No.	Number of hours	
	Shop A	Shop B
115	25	25
118	90	30
120	75	10
121	65	--
124	<u>20</u>	<u>10</u>
	275	75

Indirect labour: Waiting for material	20	10
Machine Breakdown	10	5
Idle Time	5	6
Overtime Premium	<u>6</u>	<u>5</u>
	316	101

A shop credit slip was issued in July that material issued under Requisition No. 54 was returned back to stores as being not suitable. A material transfer note issued in July indicated that material issued under Requisition No. 55 for Job 118 was directed to Job 124.

The hourly rate in shop A per labour hour is ₹ 3 per hour while at shop B, it is ₹ 2 per hour. The factory overhead is supplied at the same rate as in June. Job 115, 118 and 120 were completed in July.

You are asked to compute the factory cost of the completed jobs. It is the practice of the management to put a 10% on the factory cost to cover administration and selling overheads and invoice the job to the customer on a total cost plus 20% basis. What would be the invoice price of these jobs?

Solution

Factory cost statement for completed jobs

Month	Job No.	Material	Direct labour	Factory OHs	Factory cost
September	115	1,325	800	640	2,765
October	115	-	125	100	225
Total		1,325	925	740	2,990
September	118	810	500	400	1,710
October	118	515	330	264	1,109
Total		1,325	830	664	2,819
September	120	765	475	380	1,620
October	120	665	245	196	1,106
Total		1,430	720	576	2,726

Invoice price of job

Job No.	115 (₹)	118 (₹)	120 (₹)
Factory cost	2,990.00	2,819.00	2,726.00
Administration and selling OHs @ 10% of factory cost	299.00	281.90	272.60
Total cost	3,289.00	3,100.90	2,998.60
Profit (20% of total cost)	657.80	620.18	599.72
Invoice Price	3,946.80	3,721.08	3,598.32

Indirect labour costs have been included in the factory overhead which has been recovered as 80% of the labour cost.

2. Batch Costing

- It is that form of specific order costing which applies where similar articles are manufactured in batches either for sale or use within the undertaking.
- Each batch of output is a cost unit and is costed separately.
- The total batch cost divided by number of units produced in a batch gives cost per unit.
- It is generally undertaken in case of pharmaceutical production, shoes, garments, etc.

3. Economic Batch Quantity (EOQ)

It is that batch size at which sum total of ordering cost and carrying cost is minimum.

$$EBQ = \sqrt{\frac{2 \times A \times S}{C}}$$

Where, A = Annual requirement of raw material

S = Set-up cost per batch

C = Carrying cost per unit per annum

$$\text{Number of set-up} = \frac{A}{\text{Batch Quantity}} \quad (\text{Always round off to next complete value})$$

$$\text{Total Set-up cost} = \text{No. of setup} \times \text{Cost per set-up}$$

$$\text{Average quantity} = \frac{\text{Batch quantity}}{2}$$

$$\text{Total carrying cost} = \text{Average quantity} \times \text{Carrying cost per unit per annum}$$

Question – 4

AUX Ltd. has an annual demand from a single customer for 60,000 Covid-19 Vaccines. The customer prefers to order in the lot of 15,000 vaccines per order. The production cost of vaccine is ₹ 5,000 per vaccine. The set-up cost per production run of Covid-19 vaccines is ₹ 4,800. The carrying cost is ₹ 12 per vaccine per month.

You are required to:

- (i) Find the most Economical Production Run
- (ii) Calculate the extra cost that company incurs due to production of 15,000 vaccines in a batch.

Solution

- (i) Annual demand = A = 60,000 vaccines

$$\text{Set-up cost per run} = S = ₹ 4,800$$

$$\text{Carrying cost per unit per annum} = C = ₹ 12 \times 12 = ₹ 144$$

$$\text{Economic Batch Quantity} = \sqrt{\frac{2 \times A \times S}{C}} = \sqrt{\frac{2 \times 60,000 \times 4,800}{144}} = 2,000 \text{ vaccines}$$

(ii)

Statement of Cost

Particulars	Batch size = 2,000 vaccines	Batch size = 15,000 vaccines
Set-up cost	$\frac{60,000}{2,000} \times 4,800 = 1,44,000$	$\frac{60,000}{15,000} \times 4,800 = 19,200$
Carrying cost	$\frac{2,000}{2} \times 144 = 1,44,000$	$\frac{15,000}{2} \times 144 = 10,80,000$
Total Cost	2,88,000	10,99,200

Extra cost = ₹ 10,99,200 – ₹ 2,88,000 = ₹ 8,11,200

Question – 5

A jobbing factory has undertaken to supply 300 pieces of a component per month for the ensuing six months. Every month a batch order is opened against which materials and labour hours are booked at actual. Overheads are levied at a rate per labour hour. The selling price contracted for is ₹ 8 per piece. From the following data calculate the cost and profit per piece of each batch order and overall position of the order for 1,800 pieces.

Month	Batch output	Material cost (₹)	Direct wages (₹)	Direct labour hours
January	310	1150	120	240
February	300	1140	140	280
March	320	1180	150	280
April	280	1130	140	270
May	300	1200	150	300
June	320	1220	160	320

The other details are:

Month	Chargeable expenses (₹)	Direct labour hours
January	12,000	4,800
February	10,560	4,400
March	12,000	5,000
April	10,580	4,600
May	13,000	5,000
June	12,000	4,800

Solution

Statement of Cost and Profit per batch

Particulars	Jan.	Feb.	March	April	May	June	Total
Batch output (in units)	310	300	320	280	300	320	1,830
Sale value (₹)	2,480	2,400	2,560	2,240	2,400	2,560	14,640
Material cost (₹)	1,150	1,140	1,180	1,130	1,200	1,220	7,020

Direct wages (₹)	120	140	150	140	150	160	860
Chargeable expenses* (₹)	589	687	687	662	736	785	4,146
Total cost (₹)	1,859	1,967	2,017	1,932	2,086	2,165	12,026
Profit per batch (₹)	621	433	543	308	314	395	2,614
Total cost per unit (₹)	6.00	6.56	6.30	6.90	6.95	6.77	6.57
Profit per unit (₹)	2.00	1.44	1.697	1.10	1.05	1.23	1.43

Overall position of the order for 1,200 units

Sales value of 1,800 units @ ₹ 8 per unit ₹ 14,400

Total cost of 1,800 units @ ₹ 6.57 per unit ₹ 11,826

Profit ₹ 2,574

*Chargeable Expenses Rate = $\frac{\text{Total Chargeable Expenses}}{\text{Total direct labour hours}} = \frac{70,140}{28,600} = ₹ 2.452448$ per labour hour

It is assumed that recovery rate is based on overall 6 months period. Other way is to compute recovery rate for each month and then compute the cost.

Question – 6

Language Achievers, a renowned institute specializing in TOEFL preparation, has secured a spacious hall for 20,000 on weekly basis with a seating capacity of 250 students. The instructor, highly qualified and experienced, is compensated generously with an honorarium of 1,500 per lecture. Additionally, he receives reimbursement for travel expenses of ₹200 per day along with refreshments costing 1,500 per week to ensure his comfort and focus during teaching sessions. Administrative and miscellaneous expenses, covering essential utilities and materials are, 500 per week. Language Achievers has meticulously planned its curriculum, scheduling batches of 2 lectures per day, 5 days a week for 30 weeks, ensuring comprehensive coverage of the TOEFL syllabus.

Required:

- Calculate the total cost per batch.
- Determine the minimum fee per student in a batch to cover costs, if the batch is fully occupied.
- Calculate the fee to be charged from each student if batch is 80% filled and institute aims to achieve a profit margin of 25% on the fee.

Solution**(i) Calculation of Total cost per batch**

Particulars	Amount (₹)
Hall Charges (₹20,000 × 30)	6,00,000
Honorarium of instructor (₹1,500 × 2 × 5 × 30)	4,50,000
Reimbursement of travel expenses (₹200 × 5 × 30)	30,000
Refreshment (₹1,500 × 30)	45,000
Administrative and miscellaneous expenses (₹500 × 30)	15,000
Total Cost	11,40,000
No. of Batches	1
Total cost per batch	11,40,000

(ii) Minimum fee per student in a batch to cover costs

$$= \frac{\text{Total cost per batch}}{\text{Students per batch}} = \frac{11,40,000}{250} = ₹4,560$$

(iii) Number of Students if batch is 80% filled

$$= 250 \text{ students} \times 80\% = 200 \text{ students}$$

Total Fee to be recovered to achieve 25% profit margin on the fee

$$= ₹11,40,000 + (₹11,40,000 \times 1/4^{\text{th}} \text{ of sales or } 1/3^{\text{rd}} \text{ of the cost}) = ₹15,20,000$$

$$\text{Fee per student} = \frac{\text{Total Fee per batch}}{\text{Students per batch}} = \frac{15,20,000}{200} = ₹7,600$$

Question – 7

Pinku Confectioners (PC) owns a bakery which is used to make bakery items like pastries, cakes and muffins. PC use to bake atleast 50 units of any item at a time. A customer has given an order for 600 cakes. To process a batch of 50 cakes, the following cost would be incurred:

Direct materials - ₹5,000

Direct wages - ₹500

Oven set- up cost - ₹750

PC absorbs production overheads at a rate of 20% of direct wages cost. 10% is added to the total production cost of each batch to allow for selling, distribution and administration overheads. PC requires a profit margin of 25% of sales value.

Required:

- (i) Determine the price to be charged for 600 cakes.
- (ii) Calculate cost and selling price per cake.
- (iii) What would be selling price per unit If the order is for 605 cakes.

Solution**Statement of cost per batch and per order**

No. of batch = 600 units ÷ 50 units = 12 batches

	Particulars	Cost per batch (₹)	Total Cost (₹)
	Direct Material Cost	5,000.00	60,000
	Direct Wages	500.00	6,000
	Oven set-up cost	750.00	9,000
	Add: Production Overheads (20% of Direct wages)	100.00	1,200
	Total Production cost	6,350.00	76,200
	Add: S&D and Administration overheads (10% of Total production cost)	635.00	7,620
	Total Cost	6,985.00	83,820
	Add: Profit (1/3 rd of total cost)	2,328.33	27,940
(i)	Sales price	9,313.33	1,11,760
	No. of units in batch	50 units	
(ii)	Cost per unit (₹6,985 ÷ 50 units)	139.70	
	Selling price per unit (9,313.33 ÷ 50 units)	186.27	

(iii) If the order is for 605 cakes, then selling price per cake would be as below:

Particulars	Total Cost (₹)
Direct Material Cost	60,500

Direct Wages	6,050
Oven set-up cost	9,750
Add: Production Overheads (20% of Direct wages)	1,210
Total Production cost	77,510
Add: S&D and Administration overheads (10% of Total production cost)	7,751
Total Cost	85,261
Add: Profit (1/3 rd of total cost)	28,420
Sales price	1,13,681
No. of units	605 units
Selling price per unit (₹ 1,13,681 ÷ 605 units)	187.90